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Original Article Title: Topology-Constrained NeuroB-Rep: Contact-Aware CAD Reconstruction from 3-D Point Clouds

To: IEEE Access Editor
Re: Response to Reviewers

Dear Editor,

Thank you for allowing a resubmission of our manuscript, with an opportunity to address the reviewers' comments. We are uploading (a) our point-by-point response to the comments (below), (b) an updated manuscript with yellow highlighting indicating changes (as "Highlighted PDF"), and (c) a clean updated manuscript without highlights ("Main Manuscript").

Best regards,
J. W. Kang et al.

Reviewer 1

Reviewer 1, Concern #1 (Clarity of Contribution vs. Point2CAD)

The distinction between the proposed method and Point2CAD is not sufficiently clear. A side-by-side comparison table or flowchart would help readers understand the novelty.

Author response: We have added Table 2 (Pipeline Comparison), which contrasts NeuroB-Rep and Point2CAD across seven design dimensions (topology handling, contact objective, gating schedule, continuity enforcement, freeform surfaces, export format, and evaluation protocol). The Introduction (Sec. I) has been restructured so that the industrial motivation paragraph precedes the contribution list, making the positioning against Point2CAD explicit from the outset.

Author action: We updated the manuscript by adding Table 2 in Sec. IV and restructuring the Introduction (Sec. I, paragraphs 3–5).

Reviewer 1, Concern #2 (Evaluation Protocol Consistency)

The evaluation protocol mixes different scales and metrics. A rationale for this hybrid approach is needed.

Author response: We have added a dedicated "Rationale for Hybrid Evaluation" paragraph in Sec. V-B explaining why contact-gap energies are evaluated in a normalized scale while geometric distances are reported in millimeters: contact tolerances are scale-invariant design intents, whereas Chamfer/Hausdorff must be in physical units to be interpretable by manufacturing engineers. We also clarify the ICP alignment step and its role in bridging the two scales.

Author action: We updated the manuscript by adding a new paragraph "Rationale for Hybrid Evaluation" in Sec. V-B.

Reviewer 1, Concern #3 (Missing Baseline Comparison)

Table 3 reports only your method. At least one baseline (e.g., Point2CAD) should be included under the same protocol.

Author response: Table 7 (Geometry Consistency after ICP) now includes a Point2CAD column. We ran Point2CAD on the same three components (Impeller, Shaft, Casing) under the identical ICP protocol with 120 independent repeats per component. NeuroB-Rep consistently outperforms Point2CAD: geometric accuracy and surface coincidence errors are 2–3× lower, while length errors show 4–8× improvement. (Note: The original Table 3 is now Table 7 due to newly added tables.)

Author action: We updated the manuscript by extending Table 7 with a Point2CAD column (Sec. VI-G).

Reviewer 1, Concern #4 (Failure Cases)

The paper does not discuss failure modes. Please analyze cases where the method does not perform well.

Author response: We have added Sec. VIII-E "Failure Modes and Mitigations" analyzing the cross-boundary misassignment observed in Figs. 5–6. The primary failure originates in the upstream PointNet segmentation stage when geometrically similar adjacent surfaces receive the same label. We describe the downstream effects (degraded data-term fit, missing contact edges) and propose three mitigations: adaptive layer thresholds, boundary-aware resampling, and multi-scale segmentation.

Author action: We updated the manuscript by adding Sec. VIII-E with failure analysis and three mitigation strategies.

Reviewer 1, Concern #5 (Abstract Wording)

"0.01 mm" in the abstract should reflect the range across components.

Author response: The value has been changed to "0.01–0.03 mm" to reflect the range across the three components (Impeller, Shaft, Casing).

Author action: We updated the manuscript by updating the abstract and Sec. VI with the correct range.

Reviewer 1, Concern #6 (Formula Notation)

The relationship between $d_{ij,k}$ and $s_{ij,k}$ after Eq. (2) is unclear.

Author response: We added a clarifying sentence after Eq. (2) linking $d_{ij,k}$ (gap distance) and $s_{ij,k}$ (projected overlap distance) and explaining their complementary roles in the contact objective.

Author action: We updated the manuscript by adding a clarifying sentence after Eq. (2) in Sec. IV-B.

Reviewer 1, Concern #7 (Figure Reference Order)

Figures are not referenced in sequential order.

Author response: All figure references have been checked and corrected to follow sequential order throughout the manuscript.

Author action: We updated the manuscript by correcting all figure cross-references throughout the manuscript.

Reviewer 2

Reviewer 2, Concern #1 (Comparison with Six Cited Methods)

The paper should compare with Li et al. 2019, ComplexGen, PrimitiveNet, Supervised Fitting, ParSeNet, and BrepGen, not just cite them.

Author response: We provide two levels of comparison. (1) Table 1 (Feature Comparison) gives a qualitative comparison across all surveyed methods on seven capability axes (topology support, freeform surfaces, STEP export, etc.). (2) Table 7 provides a full quantitative comparison with Point2CAD under identical protocol. A new Sec. VIII-B ("Scope of Baseline Comparisons") explains why the remaining five methods—Li et al. 2019 [?] (Supervised Fitting), PrimitiveNet [?], ParSeNet [?], ComplexGen [?], and BrepGen [?]—cannot be compared numerically: dataset mismatch (ABC/ShapeNet/synthetic vs. industrial scans), output-format disparity (chain complexes, latent codes vs. STEP), and evaluation-metric incompatibility (IoU/F-score vs. mm-locked Chamfer/Hausdorff + contact-gap). Point2CAD is the only candidate sharing the same pipeline architecture, publicly available code, and compatible input format.

Author action: We updated the manuscript by adding Table 1 (Sec. III), extending Table 7 (Sec. VI-G), and adding Sec. VIII-B explaining the scope of baseline comparisons.

Reviewer 2, Concern #2 (Paper Outline Missing)

Add a paper outline at the end of the Introduction.

Author response: A paper-outline paragraph has been added at the end of Sec. I, listing all sections and their content.

Author action: We updated the manuscript by adding an outline paragraph at the end of Sec. I.

Reviewer 2, Concern #3 (Motivation Paragraph Position)

The industrial motivation should appear before the contribution list, not after.

Author response: The motivation paragraph has been moved to precede the contribution list in Sec. I.

Author action: We updated the manuscript by reordering the motivation and contribution paragraphs in Sec. I.

Reviewer 2, Concern #4 & 5 (Related Work Structure and Depth)

The Related Work section has a single subsection and lacks detailed positioning against existing methods.

Author response: Sec. III (Related Work) has been restructured into four subsections: (A) Point-Cloud Segmentation, (B) Primitive Fitting and Assembly, (C) B-Rep Learning and Generative Models, and (D) Hybrid Reconstruction and Point2CAD. Each subsection discusses the strengths and limitations of prior methods and explicitly positions our approach. Table 1 provides a structured feature comparison. New references have been added: Li et al. 2019, PrimitiveNet (Huang 2021), Maurell et al. 2024.

Author action: We updated the manuscript by restructuring Sec. III into four subsections and adding Table 1.

Reviewer 2, Concern #6 (Face/Edge/Vertex Recovery from Points)

How are faces, edges, and vertices recovered from the unstructured point cloud?

Author response: Sec. IV-A now includes a dedicated subsubsection ("From Points to a Patch Graph," Sec. IV-A-2) explaining the kNN co-label adjacency mechanism: for each point, the k nearest neighbors are queried; if a neighbor carries a different patch label, an edge is recorded between the two patches. This yields the face adjacency graph whose edges define candidate contact boundaries.

Author action: We updated the manuscript by adding subsubsection Sec. IV-A-2 ("From Points to a Patch Graph").

Reviewer 2, Concern #7 (Single Subsection in Method)

Section III-A contains only one subsection, which violates standard structuring conventions.

Author response: A second subsubsection has been added to Sec. IV-A, resolving the single-subsection issue. Sec. IV-A now contains two subsubsections: IV-A-1 (Hybrid Fitting with INRs and Primitives) and IV-A-2 (From Points to a Patch Graph).

Author action: We updated the manuscript by adding subsubsection IV-A-2 to Sec. IV-A.

Reviewer 2, Concern #8 (Jang et al. Explanation)

The description of Jang et al. [12] is too brief. Expand the explanation of the mesh repair principles.

Author response: Sec. II-C now provides a three-step description of Jang et al.'s method: (i) detection of interpenetrating triangles via oriented bounding-box pre-filtering, (ii) computation of exact intersection curves between conflicting triangles, and (iii) local re-meshing by splitting edges along intersection curves and re-triangulating the affected patches. The method operates in a single pass without global re-meshing, making it suitable as an online diagnostic.

Author action: We updated the manuscript by expanding Sec. II-C from one sentence to a full paragraph.

Reviewer 2, Concern #9 (Projection Operator Clarity)

The passage describing the projection operator is hard to follow.

Author response: The projection operator description has been rewritten for clarity, with explicit notation for the nearest-point projection onto each parametric surface and the role of the projection in computing gap distances.

Author action: We updated the manuscript by rewriting the projection operator passage in Sec. IV-B.

Reviewer 2, Concern #10 ("Soft Barrier" Definition)

Define "soft barrier" at first use.

Author response: The term is now defined in Sec. II-C (Preliminaries) at its first occurrence: "a smooth penalty function $\log(1+\exp(\tau-s))$ that grows rapidly as the distance s between projected points on two patches decreases below threshold τ."

Author action: We updated the manuscript by adding the definition of "soft barrier" at first use in Sec. II-C.

Reviewer 2, Concern #11 (Inconsistent Paragraph Labels)

The "a:" style paragraph labels are inconsistent.

Author response: All paragraph labels have been unified using the standard `\paragraph{}` command throughout the manuscript.

Author action: We updated the manuscript by standardizing all paragraph labels throughout the manuscript.

Reviewer 2, Concern #12 (Fig. 3 Panel Count)

The caption references panels (a,b,c,d) but only three panels are visible.

Author response: This was a caption error. The original figure has always contained three panels. The caption has been corrected to reference (a), (b), and (c) only.

Author action: We updated the manuscript by correcting the Fig. 3 caption to reference three panels.

Reviewer 2, Concern #13 (PCA Normalization Justification)

The PCA canonical transform is used without justification. An ablation would strengthen the claim.

Author response: We conducted a PCA ablation experiment (5 seeds, 60 iterations each) comparing convergence with and without PCA normalization. Results are reported in Table 6 (Sec. VI-B). PCA normalization yields a modest but consistent improvement: Chamfer distance improves by 0.3% and Hausdorff by 1.4%, with lower variance across seeds. The primary benefit is stabilizing the initial fit and reducing sensitivity to scan orientation.

Author action: We updated the manuscript by adding Table 6 and Sec. VI-B ("Effect of PCA Canonical Transform").

Reviewer 2, Concern #14 (ISO 5459 Datum Explanation)

The reference to ISO 5459 needs more context.

Author response: Sec. V-A now includes an expanded explanation of the ISO 5459 datum framework: how datum features are established from physical surfaces, and how this relates to our evaluation's reference geometry and coordinate alignment via ICP.

Author action: We updated the manuscript by expanding the ISO 5459 explanation in Sec. V-A.

Reviewer 2, Concern #15 (GPU Implementation Details)

No information is provided about the GPU implementation.

Author response: We have added implementation details in Sec. V-E (Computational Cost) and Table 4: all optimizations run on a single NVIDIA RTX 3090 using PyTorch autograd with CUDA-batched energy evaluations. NeuroB-Rep requires $\sim$7 min / 4.8 GB per component vs. Point2CAD's $\sim$4 min / 3.2 GB. The overhead stems from the contact-gap and continuity terms, which require pairwise boundary-point queries absent in Point2CAD's distance-only objective.

Author action: We updated the manuscript by adding Sec. V-E (Computational Cost) and Table 4.

Reviewer 2, Concern #16 (Artifact URL)

Provide a public artifact URL for reproducibility.

Author response: Due to corporate confidentiality and intellectual property constraints on the core solver algorithms and the raw industrial scan data, we cannot publicly release the full optimization codebase. To support independent reproducibility of our reported results, we have prepared a public artifact at `https://github.com/JuneWooKang/TopoConstrained-NeuroBrep` containing three items: (1) **Evaluation scripts.** A standalone Python script (`eval/evaluate.py`) that computes Chamfer distance, Hausdorff distance, and Surface Coincidence from any pair of input meshes, following the protocol of Table 7 (Sec. V-B). Dependencies are limited to `numpy`, `trimesh`, and `scipy`; no proprietary packages are required. (2) **Default configuration file.** `configs/default.yaml` lists every hyperparameter reported in Table 3 (solver iterations, learning rate, gate schedule, loss weights, etc.), allowing readers to verify all experimental conditions without access to the solver code itself. (3) **Representative output meshes.** Final reconstructed PLY files for the Impeller component (NeuroB-Rep and Point2CAD), together with the reference mesh and a CSV summary of the Table 7 values, enabling end-to-end metric verification on the provided sample.

Author action: We updated the manuscript by adding the artifact URL and a structured description of its contents in Sec. VII (Reproducibility and Artifact), and releasing `configs/default.yaml` to make all reported hyperparameters independently verifiable.

Reviewer 2, Concern #17 (Table 3 Baseline Column)

Table 3 should include at least one baseline for comparison.

Author response: The consistency table (now Table 7 in the revised manuscript) now includes a Point2CAD column evaluated under the identical ICP protocol (120 repeats per component). See Concern #3 of Reviewer 1 above.

Author action: We updated the manuscript by adding a Point2CAD column to Table 7 (Sec. VI-G).

Reviewer 2, Concern #18 ("Reproducibility Hygiene" Wording)

The term "reproducibility hygiene" is informal.

Author response: The term has been replaced with "reproducibility protocol" throughout the manuscript.

Author action: We updated the manuscript by replacing "reproducibility hygiene" with "reproducibility protocol" throughout Sec. VII.

Additional Corrections (author-initiated)

During revision, we identified and corrected the following issues independently of the reviewer comments:

1. **Evaluation repeat count (1,000 → 120):** To ensure a strictly fair, apples-to-apples comparison with the newly added Point2CAD baseline, we re-evaluated both methods under the exact same protocol. Consequently, the evaluation was standardized to 120 independent repeats per component (rather than the 1,000 repeats initially reported for our method alone). The text and Table 7 caption have been updated to reflect this standardized protocol.
2. **Fig. 3 caption (4 panels → 3 panels).** The figure has always contained three panels; the caption erroneously referenced four panels (a,b,c,d). Corrected to panels (a), (b), (c).
3. **Abstract grammar.** "a based on temperature gating mechanism" corrected to "a temperature-based gating mechanism."
4. **Computational cost reporting (Table 4).** Added Table 4 (Sec. V-E) reporting training time, GPU memory, and per-component overhead for both NeuroB-Rep and Point2CAD on an NVIDIA RTX 3090.

5. **New Acknowledgment section.** Added NRF funding acknowledgment before the References, per IEEE Access policy.
6. **Section and table renumbering.** A new Sec. II (Preliminaries) was inserted and four new tables (Tables 1, 2, 4, 6) were added, causing all subsequent section and table numbers to shift. All cross-references in the revised manuscript have been updated accordingly.

We believe these revisions comprehensively address all reviewer concerns. We are grateful for the opportunity to strengthen the manuscript and welcome any further suggestions.